

Course Title: Introduction to Mechanical Engineering

Course Code: 22ESC144

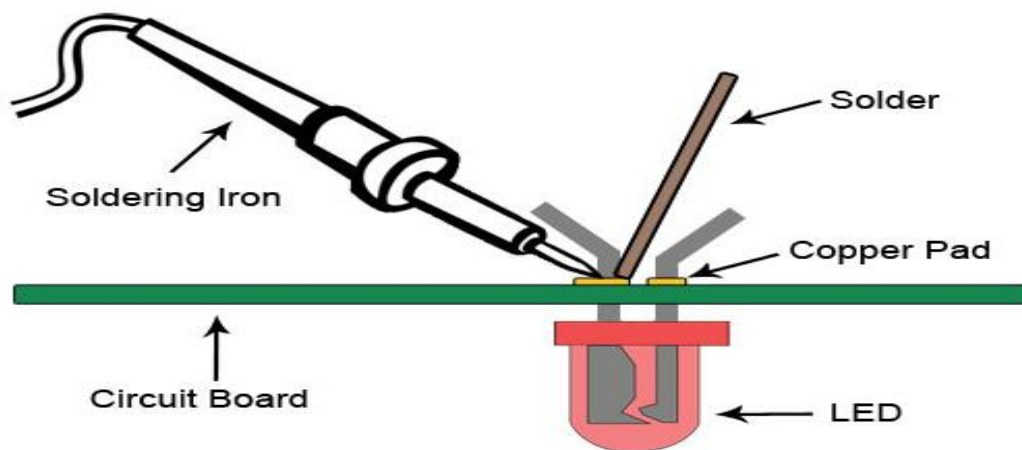
Course In charge: Ashoka E

Module-4

Joining Processes: Soldering, Brazing and Welding, Definitions, classification of welding process, Arc welding, Gas welding and types of flames.

Soldering

Soldering is the process of joining two or more electronic parts together by melting solder around the connection. The Solder or filler material is a metal alloy which is melted using a hot iron at a temperature less than 450°C and when it cools it creates a strong electrical bond between the parts. It is used to join electronic components like copper wires, printed circuit boards (PCB), etc



Soldering Tools

Soldering Iron or Torch

The Soldering Holder is the main part or device in the soldering process. It does consist of a long metallic tip. It converts the electrical energy into heat energy and helps to melt the filler material and join the two parts.

Solder or Filler Material

The Filler Material is the combination of Tin and Lead (call alloy) where the soldering holder will melt it and joins the work pieces. Filler Material is also called as a Solder.

Flux Material:

During the joining process, there is a chance of oxidization which is an avoiding chemical process while joining. It leads to the damage of the joint by creating improper joint and holes in the welding. To prevent

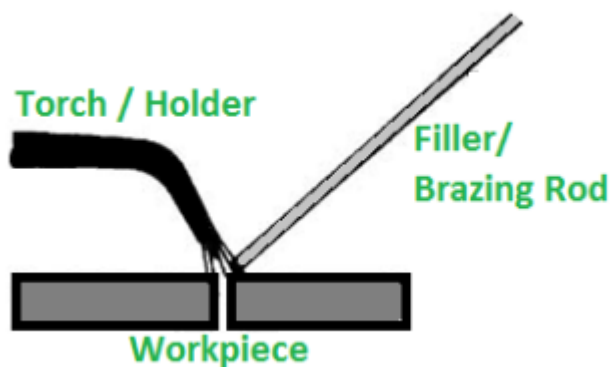
this oxidization, the flux material will be used while welding. The Flux material is a combination of Zinc Chloride, Resin, and Ammonium Chloride.

Working Method of soldering operation

Before the joining process, the surfaces of the work pieces are cleaned properly to avoid dust and improper joining. Now, the filler (Tin and Lead Alloy) is dipped in flux material to avoid oxidization during joining. The holder is heated to gain the required temperature. Now, the holder will be moved on to the work pieces along with filler material to perform the joining process. The soldering will be successfully done once the filler material is melted and allowed to join the two work pieces using the holder

Brazing:

Brazing is a joining process traditionally applied to metals (like brass, copper, aluminium in which molten filler metal (the braze rod) flows into the joint. In this process, the joining of materials is performed at a temperature more than 450°C . It is used in joining the mechanical components like brass, copper, aluminum, etc.,



The melting point of the filler metal is above 450°C , but always below the melting temperature of the parts to be joined, which distinguishes the process from welding where high temperatures are used to melt the base metals together.

The filler metal, while heated slightly above melting point, is protected by a suitable atmosphere which is often a flux. The molten filler metal cools to join the workpieces together providing a strong join between similar or dissimilar metals

To achieve a sound brazed joint, the filler and parent materials should be metallurgically compatible, and the joint design should incorporate a gap into which the molten braze filler can be drawn or distributed by capillary action.

Brazing Process:

Parts of Brazing

Torch or Holder:

The Torch or Holder is the main component where the **gas** is allowed to pass through it and at the **end of the torch**, the fire will be produced. Using this fire, the workpiece surfaces are heated at first (before joining).

Filler Material:

Similar to the soldering process, the brazing also consists of filler material which is melted by torch and allowed onto the workpieces for joining. The filler material depends upon the *type of workpiece material*

Flux Material:

The flux material is oxidization preventing paste while welding the workpieces

Oxy-Acetylene Gas:

The brazing process is not at all an **electrical component** where the heat can generate using current. It uses the oxy-acetylene gas to produce the fire for melting the filler material and joining the work pieces.

Brazing Operation

In brazing, the filler metal has a lower melting point than the parent metal. So first of all, this filler metal is heated and as a consequence of this heating, this filler metal starts to melt. This filler metal then allowed to flow into the joint formed by the work pieces. At the same time this filler metal is protected by a flux. As this filler metal is in liquid form starts to flow over parent metals (wetting takes place) and it is then slightly cooled to form the solid and clear joint.

Difference between Soldering and Brazing

The difference between soldering and brazing process according to their **Working and Operation** are

Soldering Process	Brazing Process
1. The temperature range used for Soldering is Less than 450 ⁰ C degrees centigrade which is less than brazing temperature.	1. The temperature range used for Brazing is more than 450 ⁰ C centigrade
2. It uses Electrical Energy to produce heat.	2. Brazing uses gas and fire to produce heat.

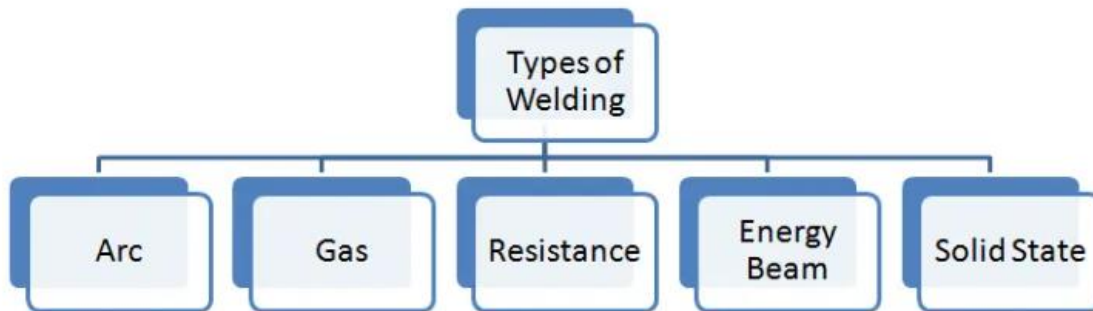
3. Uses Filler Material and Flux material	3. Uses Filler Material and Flux material
4. The Flux material is made up of Tin and Lead.	4. The Flux material is made up of Copper and Zinc.
5. No preheating is required (<i>but we can pre-heat the soldering machine for better quality</i>)	5. Pre-heating of the brazing machine is required to acquire the joining temperature.
6. Soldering can be operated easily and no technical man-power is required	6. Brazing can be a little harder to operate compare to soldering and it requires skilled labour
7. The soldering is cheaper in use and the soldering operation is faster.	7. Even the brazing operation is also faster but not as fast as soldering.
8. The joints are temporary which is easy to remove and weak in nature to brazing joints.	8. The brazing joints are permanent and stronger but not as strong as the base metal (<i>the workpiece material</i>) compared to soldering joints.
9. The Soldering process is used to join the soft materials and cannot be used to join metals like cast iron, steel, etc.	9. The Brazing Process is used to join the hard materials and cannot be applied on the metals like copper wires, electronic parts, etc.
10. It is used in electronic devices.	10. It is used in refrigeration parts and to join the machine tools components, machines

Welding

It is a joining process of metals with the help of high heat that melts the metals and fuse them together to form a strong weld joint on cooling. Pressure is also applied in some cases to make the metals to fuse together during the cooling of the weld. In some cases, Filler metal is used for joining the metals pieces together. Unlike brazing and soldering in which the base metal is not melted, the base metal melts in the welding process.

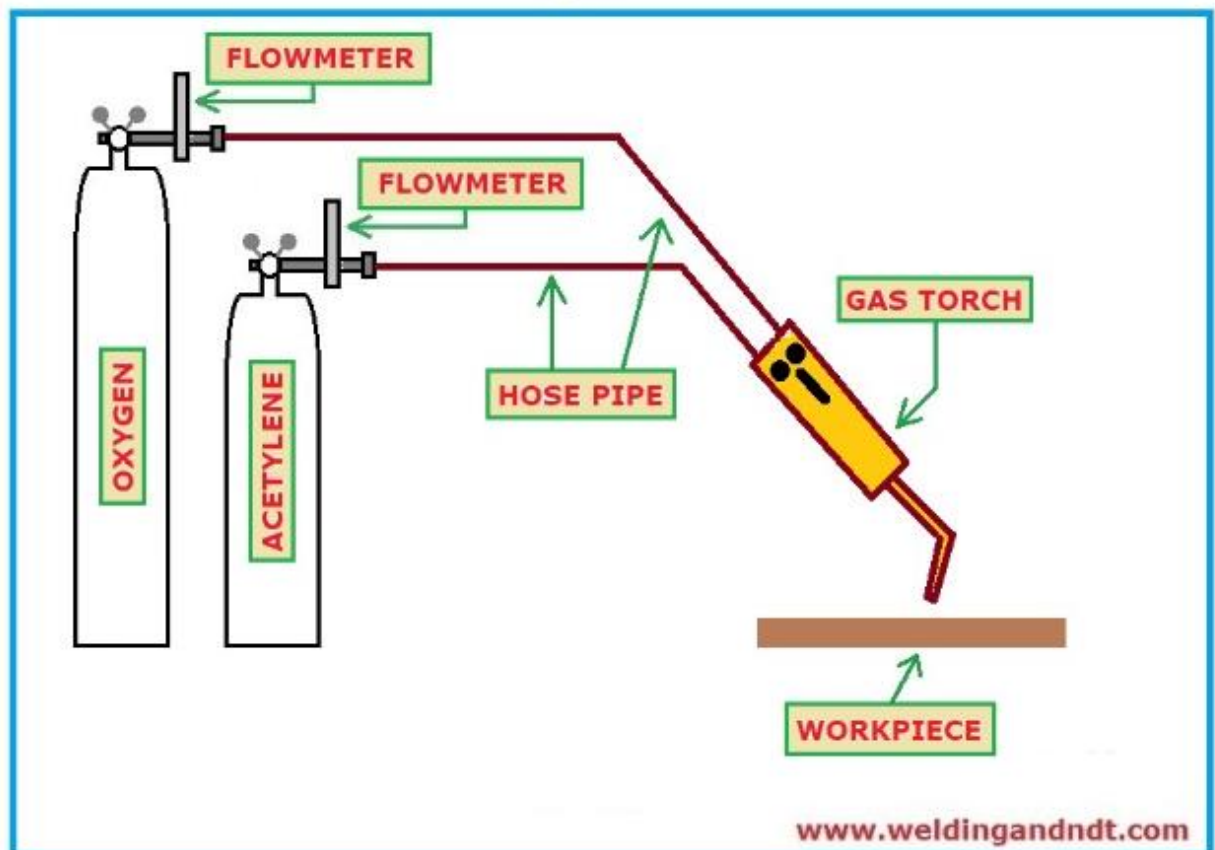
Classification of Welding

Welding can be classified mainly into 5 types and these are:



Gas or Oxy Acetylene Welding

Oxy Acetylene Welding is a type of **Gas Welding Process** that is used to join and cut the metals. In the name Oxy Acetylene Welding, we can observe two words **oxy** and **acetylene**; the word **Oxy** meant **Oxygen**, and **Acetylene** is the **order-less hydrocarbon gas**. So, the combination of oxygen and acetylene both are used to weld the metal parts. It is used for joining most ferrous and non-ferrous metals, carbon steels, alloy steels, cast iron, aluminum and its alloys, nickel, magnesium, copper and its alloys, etc



Parts of Oxy Acetylene Welding

- **Gas Cylinders:** In the Gas Cylinders, the gases like Oxygen and Acetylene are stored at high pressure whenever the welding needs to be done, the gas is released manually.
- **Regulator/ Flow Meter:** A Regulator is a fixed part of the Cylinder; it is used to stop or slow down the pressure of the gas that is used to weld
- **Welding Torch/ Gas Torch:** A Welding Torch is the main component in the oxy acetylene equipment where the flame is produced and used to weld or cut the materials. A welding torch produces a high-temperature flame according to the welding material.
- **Non-Return Valve/ Hose Pipe:** Non-Return Valve is a type of valve that ensures the flow of gas is in one direction (that too out of the cylinder, regulator, and to the Torch.)

Working of Oxy Acetylene Welding

In this process, the **welding torch** produces the **high-temperature flame about 3500⁰C** with the help of oxygen and acetylene. Due to the high flame, the metal that needs to be weld gets weak and melted, meanwhile, a **filler material** is used to fill the gaps between two workpieces. As the filler is cooled, the two workpieces will be joined.

Advantages and Disadvantages of Oxy Acetylene Welding

The advantages of Oxy Acetylene Welding are

- It is **simple** in construction, no huge parts and equipment are used in its installation like Arc Welding.
- Not only for welding but also oxy acetylene is used to separate or **cut** the materials into pieces.
- Equipment is cheaper than other welding processes, easy to learn and use
- Since no electricity is required, hence can be used at locations where power sources are not available
- The intensity of the flame can be adjusted to high and less by using the valve.
- It can be used for high melting metals and even for low melting metals.

The disadvantages of Oxy Acetylene Welding are:

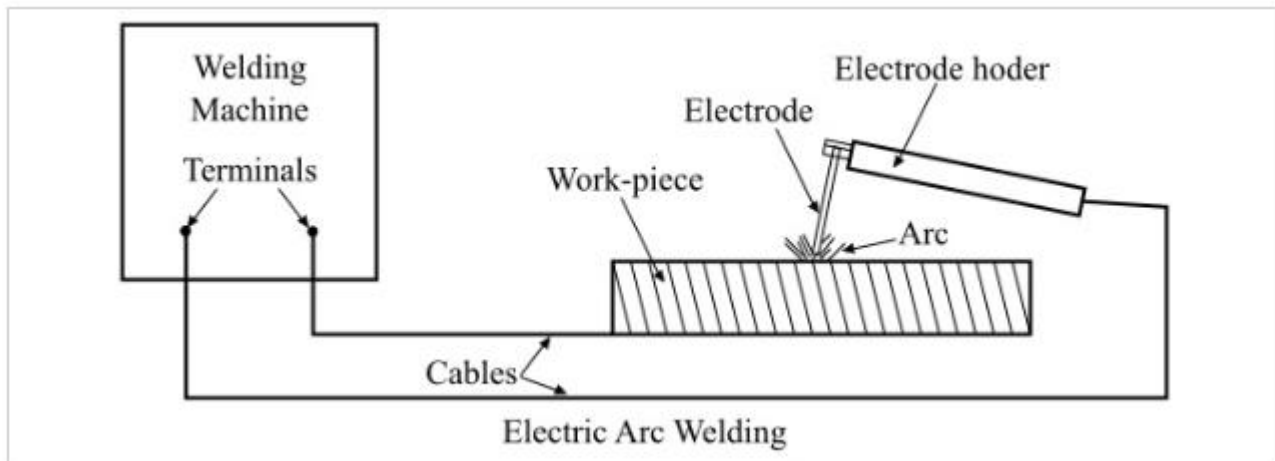
- The temperature of the flame is **less** when compared to the arc welding.
- Slow welding if compared to other arc welding processes, hence less productive
- Due to the presence of combustible gas (Acetylene), lots of Safety precautions needed because these gases are highly flammable and can explode if catches fire

Electric Arc Welding

The process of welding in which heat is produced by creating an electric arc to join metal workpieces is known as **electric arc welding**. It's the most common industrial welding process today, and can be used for steel, cast iron, magnesium, and many other metals

Electric arc welding is a type of welding that uses a welding power supply to create an electric arc between a metal stick, called *electrode*, and the workpiece to melt the metals at the point of contact

A simplified circuit of electric arc welding is shown in the figure.



In electric arc welding, either AC or DC current is obtained from a welding power supply. Here, one terminal is connected to the electrode mounted on an electrode holder, which is held by the welder, while the other terminal is connected to the workpiece and the circuit is completed through an air gap between the electrode and the work-piece.

The length of the air gap (i.e., distance between electrode tip and the surface of the work-piece) is about 3 mm to 6 mm. The welding is done by creating an electric arc between the electrode and the workpiece. The temperature of the arc is very high (about 3500 °C to 4000 °C) and the metal in contact with the arc becomes molten which enables a weld to be made. The electrode is then moved slowly in the desired direction to complete the weld.

Working of Electric Arc Welding

The process of electric arc welding is based on the principle that, when electric current is passed through an air gap from one electric conductor to another, then an electric arc is produced which generates a very intense and concentrated heat.

The temperature of the arc between two conductors is approximately 3500 °C to 4000 °C. This high temperature generates intense heat in the arc at the point of welding, which melts a small portion of metal in the work-piece.

The electric arc keeps this molten metal pool agitated and the base metal is thoroughly mixed with melted electrode metal, after that the metal pool cools down under a protective cover of slag left by the electrode. On cooling, a strong weld joint is formed between the two metal pieces.

Advantages of Arc Welding

Some of the chief advantages of the electric arc welding are given as follows –

- The electric arc welding is the suitable welding process for high speed welds.
- Apparatus required for arc welding is very simple and portable.
- The electric arc welding gives superior temperature at the point of welding.
- Electric arc welding can work on both AC and DC supply.
- It is inexpensive to install.

Disadvantages of Electric Arc Welding

The disadvantages of electric arc welding are as follows –

- The welding process with electric arc welding requires skilled operators.
- Electric arc welding cannot be used for welding of reactive metals such as aluminium, titanium, etc.
- Electric arc welding is not suitable for welding thin metals.

Applications of Electric Arc Welding

The important applications of electric arc welding are as follows –

- Electric arc welding is used in repairing of broken parts of machines.
- It is used for welding of cast iron or steel housings and frames.
- Electric arc welding is used in various industries such as automotive industries, construction industries, mechanical industries, etc.

Difference between Welding, Brazing, and Soldering

Method	Welding	Brazing	Soldering
Strength of the Joint	These are the strongest joints that can bear the load.	The strength at the joint soldering but weaker than welding.	The strength at the joint is very less.
Heating Temp	The desired welding area temperature is up to 3800 °C.	In brazing, it can go up to 600°C.	The temperature required is up to 450 °C.

Melting point	Joining requires the work pieces to be heated to their melting point.	The work piece is heated but below the melting point.	The work piece does not need to be heated.
Cost	The setup cost is high compared to brazing and soldering.	The setup cost is high compared to brazing.	The setup cost is very small compared to brazing and soldering.
Filler	Filler Material Optional.	Filler materials are essential.	Filler material mandatory.
Flux	Flux use is optional.	Flux required	Flux required
Easy of Joining the material	Difficult to weld different metals.	Different metals can be joined.	Different metals can be joined.
Difficulty in making thin sheets	Difficult to weld thin sheets.	Thin sheets can be joined.	Thin sheets can be joined.
Skill Required	Involving heat consumption requires high-level skill.	The costs involved and the skills required are among soldering and welding.	The cost involved and the skill required are very low.
Effect on Mechanical Properties	The <u>mechanical properties</u> of the base metal may vary in the joint space due to heating and cooling.	The mechanical properties of the joint may change, but this is almost negligible.	There is no change in the mechanical properties after joining.
Preheating	Since this is done at high temperatures, it is not necessary to preheat the work before welding.	Preheating is beneficial for forming solid joints because brazing is performed at relatively low temperatures	No preheating is required.

Difference between Arc Welding and Gas Welding

Sl.NO	Arc Welding	Gas Welding
1	In the arc welding process, an electric arc is created to produce the required heat for fusing the base metal	In gas welding, combustion of gaseous fuel with oxygen is used to generate the required heat
2	Arc welding can only be used for the metals which are electrically conductive.	Gas welding is equally applicable for both electrically conductive and non-conductive metals.
3	Electric arc welding requires availability of electric power supply for its operation.	No electric power supply is required in case of gas welding.
4	In case of arc welding, no cylinder containing fuel and oxygen is required.	In case of gas welding, two cylinders are employed, one for gaseous fuel such as acetylene, propylene, etc. While another is required for containing oxygen.
5	Arc welding can be used for joining thicker components.	Gas welding is suitable only for thin plates and sheet metals.
6	Arc welding is not suitable for heating, cleaning, cutting, etc.	In addition to the joining of metals, the gas welding can be used for heating, gas cutting, brazing, soldering and cleaning, etc.
7	As the arc temperature is very high, the temperature of the core is about more than 6000 °C. Therefore, arc welding can be used for joining of metals having high melting point..	Temperature of the gas flame is comparatively low, about 3500 °C. Therefore, gas welding cannot be used for welding of the metals having very melting point.
8	The electric arc has a narrow cross-sectional area. Therefore, it reduces the width of weld bead and heat affected zone.	The area of cross-section of the gas flame is relatively wider. Therefore, it results in wider weld bead and heat affected zone.

Types of oxy-acetylene flames

The types of flames depend on the gas ratio i.e. ratio of the parts of oxygen to the parts of the acetylene. Depending on the gas ratio following flames are obtained

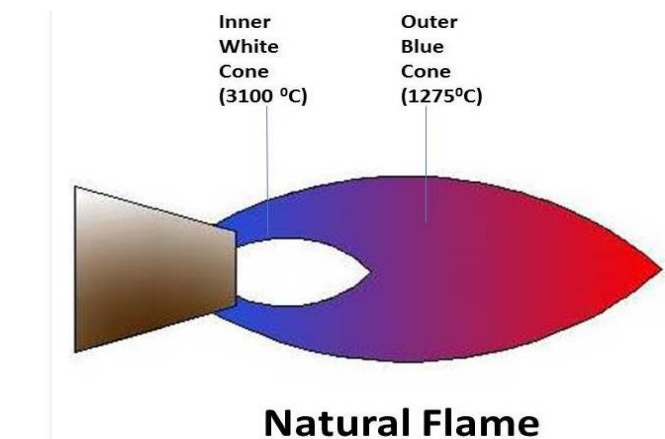
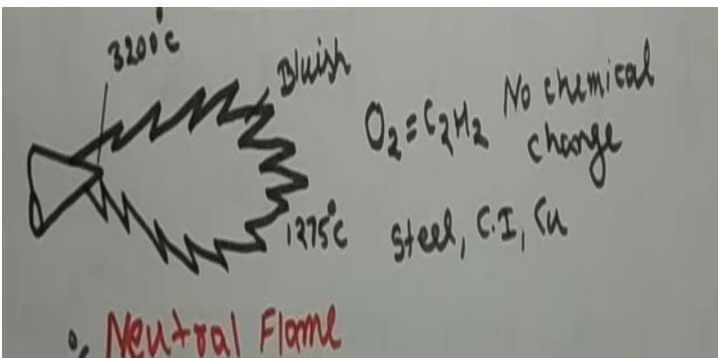
- (i) Neutral flame
- (ii) Oxidizing flame
- (iii) Reducing flame (Carburizing flame)

In this welding process gases fuels are burn and produce high temperature flames which further used to create weld joint. The flame plays lead role to create weld joint and the weld properties are highly depended on it.

(i) Neutral flame

As the name implies, this flame has equal amount of oxygen and gases fuel by the volume. This flame burns fuel completely and does not produce any chemical effect on metal to be welded. It is used for welding mild steel, stainless steel, cast iron etc. It produces little smoke.

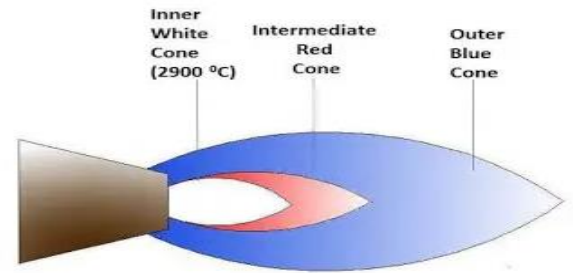
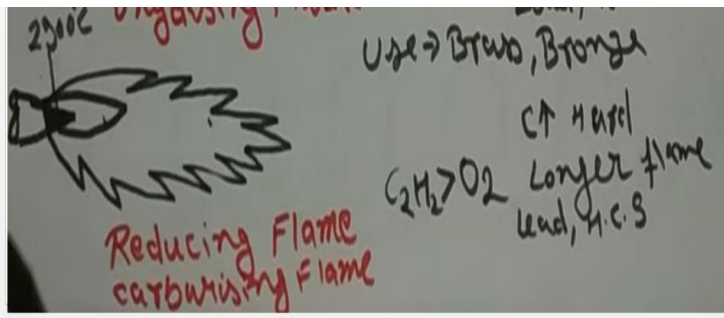
This flame has two zones. The inner zone has white in color and has temperature about 3100 to 3200°C and outer zone has blue color and have temperature about 1275°C.



ii) Carburizing Flame:

This flame has excess of fuel gas. This flame chemically reacts with metal and form metal carbide. Due to this reason, this flame does not used with metal which absorb carbon. It is smoky and quiet flame.

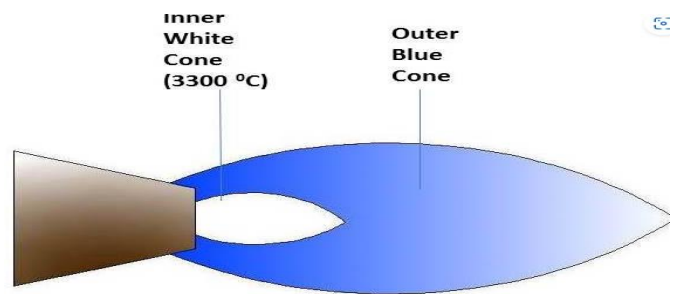
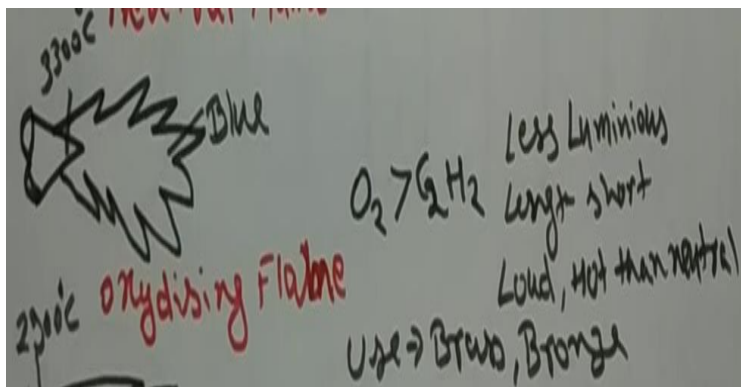
This flame has three regions. The inner zone has white color, the intermediate zone which is red in color and outer cone has blue color. The inner cone temperature is about 2900°C. This flame is used to weld medium carbon steel, nickel etc.



Carburizing Flame

iii) Oxidizing Flame: When the amount of acetylene reduces or amount of oxygen increases, the inner cone tend to disappear and the flame obtain is known as oxidizing flame. It is hotter than natural flame and has clearly defined two zones.

The inner zone has very bright white color and has temperature of about 3300⁰C.. The outer flame has blue in color. This flame is used to weld oxygen free copper alloy like brass, bronze etc.



Oxidizing Flame

Course In charge: Ashoka E

Course Title: Introduction to Mechanical Engineering

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Course In charge: Ashoka. E

Engineering Materials: Types and applications of Ferrous & Nonferrous Metals, silica, ceramics, glass, graphite, diamond and polymer. Shape Memory Alloys

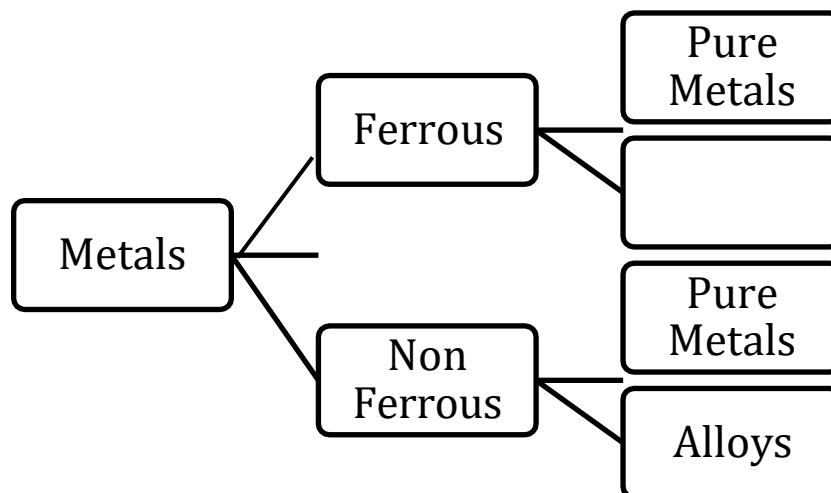
Metals

A metal is a material that is typically hard, shiny and features good electrical and thermal conductivity. Metals are generally malleable: they can be hammered or pressed permanently out of shape without breaking

Metals can be either ferrous or non-ferrous. Ferrous metals contain iron while non-ferrous metals do not. Both ferrous and non-ferrous metals are divided into pure metals and alloys.

A pure metal is an element – Ex: iron, copper, gold - unalloyed (not mixed) with another substance.

An alloy is a mixture of two or more elements (Ex: iron and carbon) to make another metal with particular properties (Ex: steel).



Ferrous metals

Ferrous metals contain iron. Examples are cast iron, mild steel, medium carbon steel, high carbon steel, stainless steel, and high speed steel or Tool steel.

Non-ferrous metals

Non-ferrous metals do not contain iron. Some common non-ferrous metals are aluminium, copper, zinc, tin, brass (copper + zinc), and bronze (copper + tin), Gold and Silver etc.,

Type of ferrous metal

Cast iron

Composition: Alloy of iron and 2-5% carbon, 1-3% silicon and traces of magnesium, sulphur and phosphorus

Properties and characteristics: it is very strong but brittle. Cast iron has relatively low melting point, is wear resistant, possesses good fluidity, has admirable machinability and is resistant to deformation

Application: It is used to manufacture machine frames, columns, beds and plates, automotive parts such as engine block, cylinder head, gear box case and machine parts

Cast Iron is divided into following types:

- Grey Cast Iron
- White Cast Iron
- Malleable Cast Iron
- Ductile Cast Iron
- Wrought Iron

Steel

Steel is an alloy of iron and carbon which is produced either by basic oxygen steel making process or by electric arc furnace.

Steels are broadly classified into

- a) Carbon Steels
- b) Alloy Steels
- c) Tool Steels

a) Carbon steels: Carbon steels are types of steel containing primarily iron and carbon. Other elements present in small proportions are sulphur, phosphorous, manganese and magnesium. The following types of carbon steel are.

Mild Steel

Composition: It has a carbon content of 0.05 to 0.3%. The balance is iron. The most popularly used carbon steel is mild steel.

Properties and characteristics: Tough, ductile and malleable. Good tensile strength, poor resistance to corrosion

Application: General-purpose engineering material like rivets, bolts, keys plain washer boiler plate's shaft, camshafts and gear.

Medium carbon steel

Composition: It has a carbon content of 0.3 - 0.6% carbon. The remainder is iron content.

Properties and characteristics: Strong, hard and tough, with a high tensile strength, but less ductile than mild steel.

Application: It finds application in transmission shafts, springs, spring washers, crane hooks and hand tools etc.

High carbon steel

Composition: It has a carbon content of 0.6 - 1.5%. It has an iron content of 96% to 97%

Properties and characteristics: Even harder than medium carbon steel, and more brittle. Can be heat treated to make it harder and tougher

Application: Cutting tools, hammers, chisels, screw, punches, drills lathe tools, leaf springs and milling cutter

Stainless steel

Composition: Alloy of iron and carbon with 16-26% chromium, 8-22% nickel and 8% magnesium

Properties and characteristics: Hard and tough, resists wear and corrosion

Application: Cutlery, kitchen equipment, surgical equipment, chemical handling equipment's and cutlery etc.

High speed steel or Tool Steel

Composition: Alloy of iron and 0.8 - 1.2% carbon (medium carbon steel) with tungsten, chromium, vanadium, and sometimes cobalt

Properties and characteristics: Very hard, high abrasion- and heat resistance

Application: Cutting tools

Type of Non Ferrous metal

Aluminium

Composition: Pure aluminium

Properties and characteristics: Good strength-to-weight ratio, light, soft, ductile, good conductor of heat and electricity

Application: Kitchen equipment, window frames, general cast components

Copper

Composition: Pure copper

Properties and characteristics: Malleable and ductile, good conductor of heat and electricity, good corrosion resistance and light weight.

Application: copper tubes used in refrigerator and air conditioners and radiators due to high thermal conductivity, electrical wires and cables, used to make door knobs.

Gold (Au)

Composition: Gold 91.67% Silver 5% Copper 2% Zinc 1.33% [Yellow Gold (22K)]

Properties and characteristics: It is metallic, having a yellow color. It is soft, ductile and the most malleable of metals because of this is usually alloyed to give improved strength and durability. Gold is highly reflective of heat and light. Gold is a good conductor of heat and electricity

Application: The primary application of gold is in the jewelry industry, it is an excellent conductor of heat and electricity, gold is used extensively in the electronics industries in items such as printed circuits and in transistors. The circuitry of computers, calculators, televisions, telephones, are often plated with gold.

Silver (Ag)

Composition: Most silver is a by-product of copper, gold, lead, and zinc refining.

Properties and characteristics: Silver metal in its pure state has a brilliant white metallic luster. It is a little harder than gold and is very ductile and malleable. It has the highest electrical and heat conductivity of all metals

Application: Silver is used for jewellery, coins, silverware, electrical contacts, mirror, dental alloys

Zinc

Composition: - Zinc is a heavy, bluish white metal

Properties:

- It is a fair conductor of electricity
- It has relatively low melting and boiling point
- It is resistant to corrosion

Application; - Usually used for coating steel to make galvanised items

Tin

Composition: - it is a silvery-white metal obtained from an oxide called tin stone

Properties:

- Soft, malleable and ductile
- It is corrosion resistant from water
- It has low melting point (232°C)

Application:

- It is coated for storing food and water
- Used in candles shields and mirror frames.

Alloys of copper are Brass and Bronze.

Brass

Composition: Alloy of copper (51% to 81%) and zinc (19% to 49%).

Properties and characteristics: Resistant to corrosion, fairly hard, good conductor of heat and electricity

Application: Electrical fuses and fittings, brazing solder, musical instruments such as horns, trumpets etc.,

Bronze

Composition: Alloy of copper and tin

Properties and characteristics: Fairly strong, malleable and ductile when soft

Application: Decorative goods, architectural fittings

Silica

Properties

Silica is the most abundant mineral in the earth's crust. Silica under normal conditions of temperature & pressure is a solid, crystallized mineral. It is relatively hard.

Silica quartz



Applications

- It is used for building constructions, as a refractory material and for manufacturing of glass.
- It is also used for making abrasive tools.

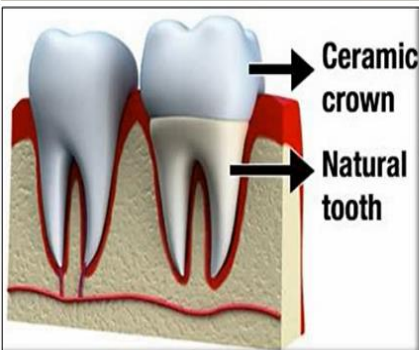
Silica sand



Ceramics

Properties

They have high melting temperature, high hardness, wear resistance. They are brittle, thermal insulators, electrical insulators, nonmagnetic and chemically stable. A ceramic is a neither metallic nor organic.



Applications

- They are used to make tiles, crockery, bricks etc.
- They are also used to make toilets, artificial bones and teeth.



Diamond

Properties

High Hardness.
Low coefficient of friction.
High thermal conductivity.
High electrical resistivity.
Low thermal expansion coefficient.
High strength.

Applications

The most familiar uses of diamonds today are: **1)** as gemstones used for adornment, **2)** as industrial abrasives for cutting hard materials **3)** as cutting tools for metal finishing process.



Glass

Properties

Glass is a non-crystalline, often transparent amorphous solid.
They are hard, brittle, smooth, resistance against abrasion and chemical attacks etc.

Applications

- 1) Packaging (jars for food, bottles for drinks)**
- 2) Tableware (drinking glasses, plate, cups, bowls)**
- 3) Housing and buildings (windows, insulation, stairs)**
- 4) Interior design and furniture (mirrors, partitions, tables)**
- 5) It is also used to make lenses of spectacles, telescope etc.**



Activate Windows
Go to Settings to activate Windows.

Graphite

Properties

It is a crystalline form of carbon. It occurs naturally and is the most stable form of carbon under standard conditions. The material is typically grayish-black in color, opaque, soft and has a radiant black sheen. Graphite has high electrical and thermal conductivity.



Applications

Graphite is used in pencils and lubricants.

Due to high electrical conductivity, it is useful in electronic products such as electrodes, batteries, and solar panels.

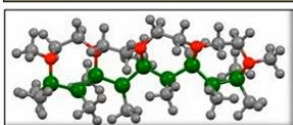


Polymers

Properties

A polymer is a large molecule or a macromolecule which essentially is a combination of many subunits.

Some polymers bend and stretch, like rubber and polyester. Others are hard and tough, like epoxies.



Applications

<i>Polymer</i>	<i>Uses of Polymer</i>
Rubber	Making tyres, elastic materials
Teflon	Non-stick cookware, plastics
Bakelite	Plastic switches, Mugs, buckets
PVC	Tubes, Pipes
Melamine	Ceramic plastic material
Nylon-6	Fabric



POLYMERS ARE EVERYWHERE



Shape memory alloys

This alloy is referred to as a smart alloy that can be deformed and returns to its pre-deformed shape when subjected to temperature change. In general, Shape memory alloys (SMAs) can memorize their earlier form when triggered by a certain stimulus like thermo mechanical or magnetic variation. This alloy is also called to be memory alloy or smart alloy or muscle wire.

Shape memory alloys (SMAs) are materials that can be deformed at one temperature but when heated or cooled, return to their original shape, i.e. the alloy appears to have a memory. Additionally, SMAs are extremely elastic (that means they can bend far and return to their original shape, like a rubber band). SMAs have this property because of two unique intermetallic phases. We call these “austenite” and “martensite,” Martensite is the low temperature phase. Austenite is the high temperature phase. These alloys are bio-compatible, lightweight, and also possess a high force-to-weight ratio

Current shape memory alloys are:

NiTi, or nitinol. This is the most developed SMA, with excellent mechanical properties. It is pretty much the only commercially viable SMA.

Cu-Al-Ni, Fe-Mn-Si, and Cu-Zn-Al systems. These have poorer mechanical properties and are still in development.

Application of Shape memory alloy

One of the most interesting potential applications for shape memory alloys is in prosthetic, robotic hands. Recent research of SMA has been applied in the field of Aerospace, Automotive, Biomedical, Orthodontic field and Robotics.

What is the Shape Memory Effect (Optional)

At last, the magical effect that gives SMAs their name the shape memory effect (SME) is what allows SMAs to return to their original shape after heating. The SME is a temperature-induced martensite-to-austenite phase transformation.

